Comprehensive Analysis of E-Commerce Transactional and Customer Data

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Project Overview

This project analyzes two connected datasets from an e-commerce company:
 customer data and transaction data.
 The goal is to understand customer behavior, sales patterns, and profitability
 using data analysis and machine learning techniques.

Data Description

The analysis is based on two datasets:

- Transactions dataset: 10,486 records
- Customers dataset: about 1,500 records

Transaction data includes order dates, revenue, profit, discounts, and locations.
Customer data includes age, income, job role, department, and satisfaction indicators.

	transaction_id	zip_code	country_code	state_name	city_name	delivery_type	sales_zone	client_segment	client_id
0	T01936	85023.0	USA	Arizona	Phoenix	Standard Class	West	Consumer	C0568
1	T00178	98198.0	USA	Washington	Des Moines	Second Class	West	Consumer	C1308
2	T04512	NaN	USA	Illinois	Quincy	Standard Class	Central	Consumer	C0475
3	T01059	78041.0	USA	Texas	Laredo	Standard Class	Central	Consumer	C0445
4	T04032	92037.0	USA	California	San Diego	Standard Class	West	CONSUMER	C0462

	client_id	Age	Attrition	BusinessTravel	DailyRate	Department	DistanceFromHome	Education	Ed
0	C0395	43	No	Travel_Frequently	313	Research & Development	21.0	3.0	
1	C0840	35	No	Travel_Rarely	982	Research & Development	1.0	4.0	
2	C1230	24	No	Travel_Rarely	506	Research & Development	29.0	1.0	
3	C0507	29	No	Travel_Rarely	1176	Sales	3.0	2.0	
4	C0853	19	No	Travel_Rarely	645	Research & Development	9.0	2.0	

Data Cleaning & Preparation

Before analysis, a full data cleaning process was performed:

- Missing numerical values were filled using the median
- Missing categorical values were filled using the most frequent value
- Incorrect data types were fixed (dates and numeric fields)
- Outliers were detected using the IQR method and capped
- Categorical variables were standardized (lowercase, trimmed)
- Logical errors were corrected (negative values, profit greater than revenue)

This ensured high data quality and reliable results.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 10486 entries, 0 to 10485
Data columns (total 17 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   transaction_id    10486 non-null  object
 1   zip_code          10174 non-null  float64
 2   country_code      10486 non-null  object
 3   state_name        10486 non-null  object
 4   city_name         10486 non-null  object
 5   delivery_type     10173 non-null  object
 6   sales_zone        10486 non-null  object
 7   client_segment    10486 non-null  object
 8   client_id         10486 non-null  object
 9   order_ts          10486 non-null  datetime64[ns]
 10  dispatch_ts       10276 non-null  datetime64[ns]
 11  item_id           10486 non-null  object
 12  item_description  10486 non-null  object
 13  units_count       10486 non-null  int64
 14  total_sum         10171 non-null  float64
 15  profit            10169 non-null  float64
 16  discount          10176 non-null  float64
dtypes: datetime64[ns](2), float64(4), int64(1), object(10)
memory usage: 1.4+ MB
None
```

```
Missing values:
 transaction_id        0
zip_code             312
country_code           0
state_name             0
city_name              0
delivery_type        313
sales_zone             0
client_segment         0
client_id              0
order_ts               0
dispatch_ts          210
item_id                0
item_description       0
units_count            0
total_sum            315
profit               317
discount             310
dtype: int64

Duplicates: 0
```

```
Data columns (total 36 columns):
 #   Column                    Non-Null Count  Dtype
---  ------                    --------------  -----
 0   client_id                 1495 non-null   object
 1   Age                       1495 non-null   int64
 2   Attrition                 1495 non-null   object
 3   BusinessTravel            1495 non-null   object
 4   DailyRate                 1495 non-null   int64
 5   Department                1495 non-null   object
 6   DistanceFromHome          1377 non-null   float64
 7   Education                 1376 non-null   float64
 8   EducationField            1495 non-null   object
 9   EmployeeCount             1495 non-null   int64
 10  EmployeeNumber            1495 non-null   int64
 11  EnvironmentSatisfaction   1495 non-null   int64
 12  Gender                    1495 non-null   object
 13  HourlyRate                1495 non-null   int64
 14  JobInvolvement            1495 non-null   int64
 15  JobLevel                  1495 non-null   int64
 16  JobRole                   1495 non-null   object
 17  JobSatisfaction           1376 non-null   float64
 18  MaritalStatus             1495 non-null   object
 19  MonthlyIncome             1377 non-null   float64
 20  MonthlyRate               1495 non-null   int64
 21  NumCompaniesWorked        1495 non-null   int64
 22  Over18                    1495 non-null   object
 23  OverTime                  1495 non-null   object
 24  PercentSalaryHike         1495 non-null   int64
 25  PerformanceRating         1495 non-null   int64
 26  RelationshipSatisfaction  1495 non-null   int64
 27  StandardHours             1495 non-null   int64
 28  StockOptionLevel          1495 non-null   int64
 29  TotalWorkingYears         1495 non-null   object
 30  TrainingTimesLastYear     1495 non-null   int64
 31  WorkLifeBalance           1495 non-null   int64
 32  YearsAtCompany            1495 non-null   int64
 33  YearsInCurrentRole        1495 non-null   int64
 34  YearsSinceLastPromotion   1495 non-null   int64
 35  YearsWithCurrManager      1495 non-null   int64
dtypes: float64(4), int64(21), object(11)
memory usage: 420.6+ KB
None
```

```
Age    DailyRate  DistanceFromHome    Education
```

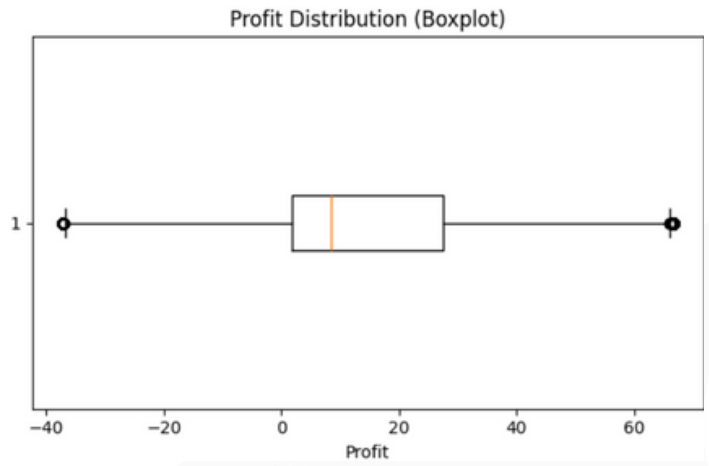
Exploratory Data Analysis (EDA)

The dataset contains 10,486 transactions.
Most orders include 2–5 items, with an average of 3.7 items per order.
The average order value is 140 USD, while 50% of orders are below 55.5 USD, indicating a strongly right-skewed revenue distribution.
The mean profit is 14.8 USD, but some transactions generate losses (minimum profit −37 USD).
Discounts are generally low: at least 75% of orders have no discount, and the maximum discount is 50%.

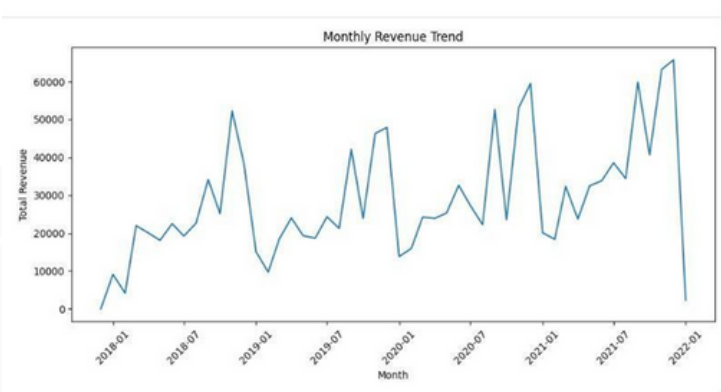
	units_count	total_sum	profit	discount
mean	3.733931	140.315662	14.806053	0.142401
median	3.000000	55.500000	8.386800	0.200000
std	2.070196	168.686898	28.093906	0.158842

	units_count	total_sum	profit	discount
count	10486.000000	10486.000000	10486.000000	10486.000000
mean	3.733931	140.315662	14.806053	0.142401
std	2.070196	168.686898	28.093906	0.158842
min	1.000000	0.444000	-37.229837	0.000000
25%	2.000000	17.920000	1.734300	0.000000
50%	3.000000	55.500000	8.386800	0.200000
75%	5.000000	209.692500	27.717600	0.200000
max	9.000000	497.351250	66.686062	0.500000

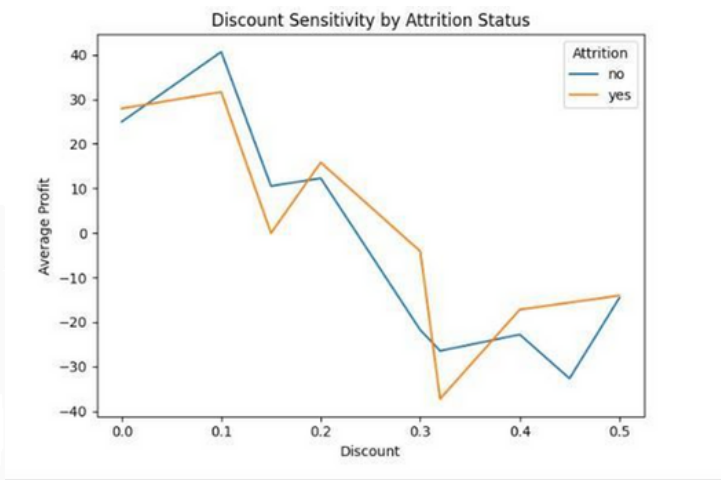
Exploratory Data Analysis (EDA)



The profit boxplot shows that while most transactions are profitable, negative-profit outliers exist, confirming that not all sales generate value.



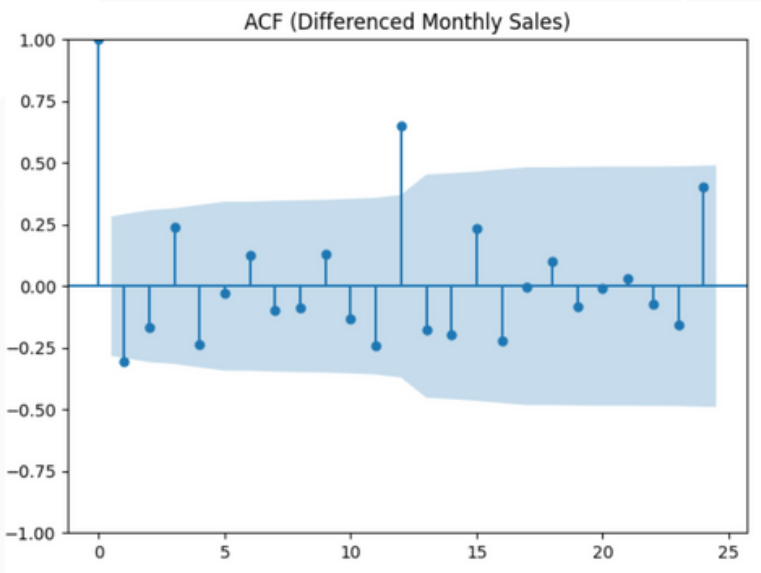
The monthly revenue trend shows clear fluctuations over time, with an overall upward pattern and several sharp peaks.
These peaks may be related to promotions, seasonal demand, or special sales events.

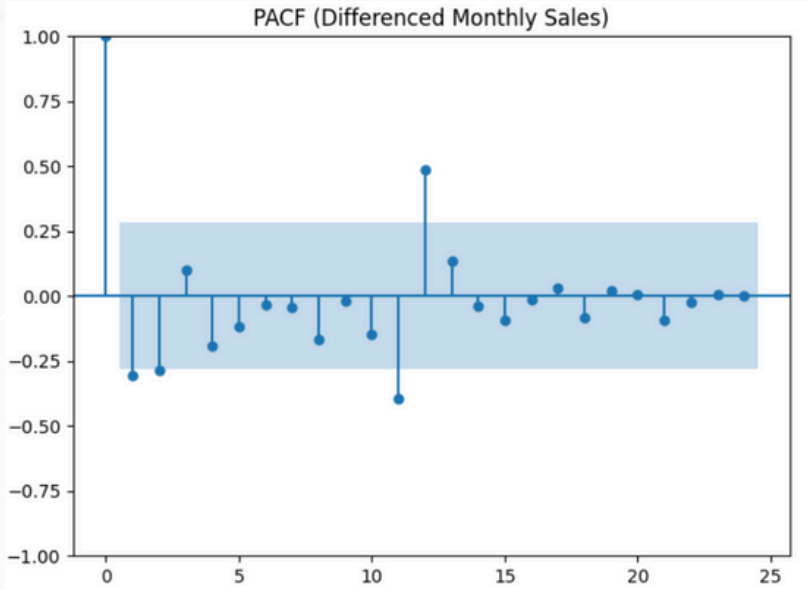


The discount sensitivity chart demonstrates that higher discounts lead to lower average profit for both retained and attrited customers.
Large discounts often result in negative profit and do not significantly improve customer retention.

ACF and PACF Analysis

The original monthly sales series is non-stationary (ADF p-value ≈ 0.99, fail to reject H_0).
After applying first-order differencing, the series becomes stationary (ADF p-value < 0.001, reject H_0).
The ACF and PACF plots of the differenced series show that most lags lie within the confidence bounds, with a few significant spikes.
This indicates limited autocorrelation and supports the use of a SARIMA model with low-order AR and MA terms.





Time Series Modeling & Forecasting

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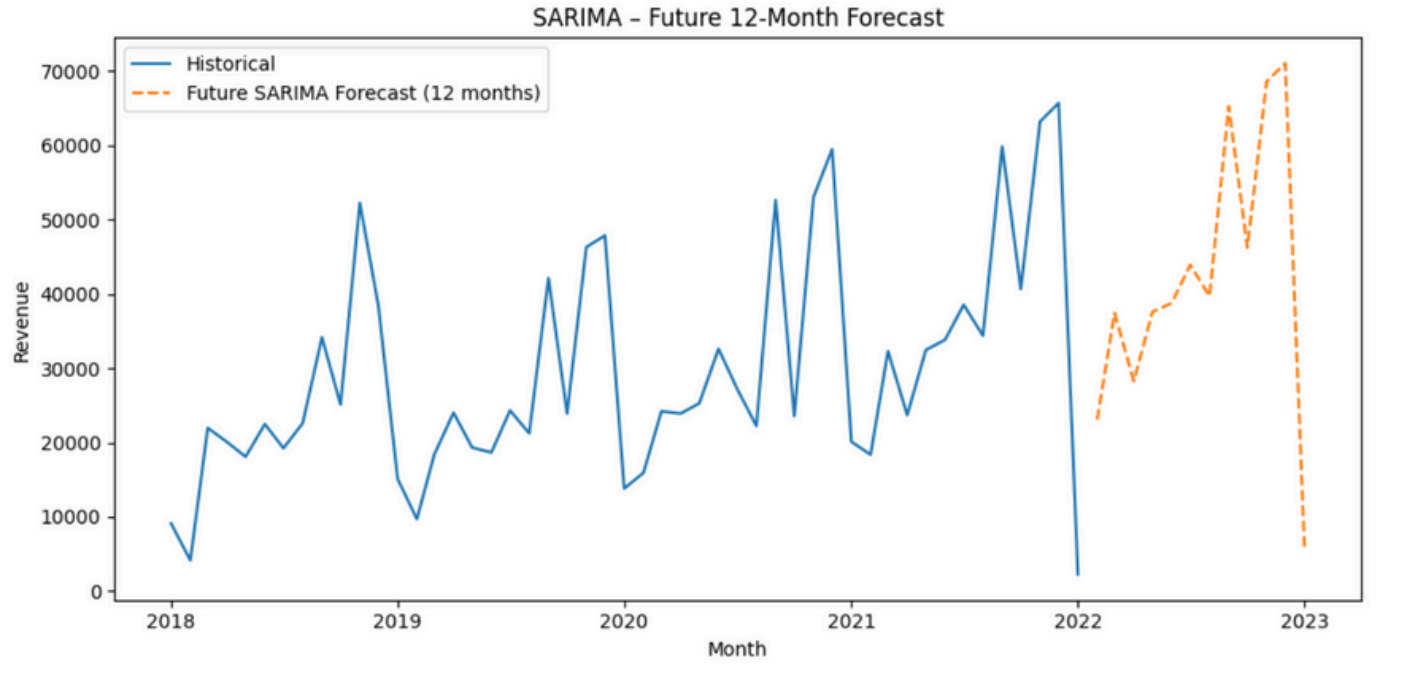
Forecast accuracy on last 12 months:
RMSE: 8540.129785371997
MAE : 5868.52428730966
```



A SARIMA model was applied to monthly revenue data.
The model captures trend and seasonality and was used to forecast revenue for the next 12 months.
Forecast accuracy is moderate due to limited historical data.

Customer Segmentation (Clustering)

Customers were segmented using K-Means clustering. Transaction data was aggregated at customer level
and combined with demographic features.
Two main customer groups were identified:
low-value and high-value customers.

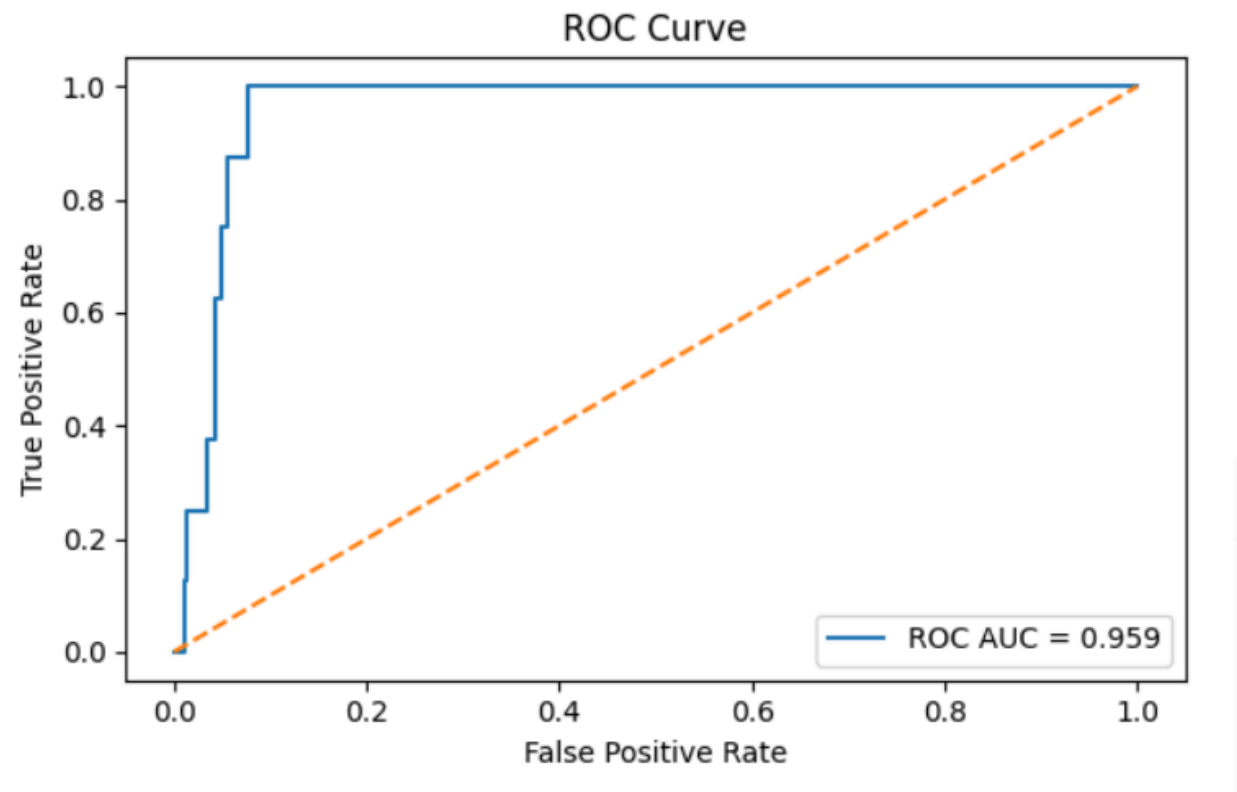
Customer Segments Interpretation

High-value customers are older, have higher income, place more orders, and generate most revenue and profit.
Low-value customers buy less frequently and generate lower profit.
This segmentation can support targeted marketing strategies.

	client_id	num_orders	total_revenue	total_profit	avg_order_value	avg_discount	total_units	unique_items
0	c0000	6	105.09000	50.161500	17.515000	0.283333	20.0	6
1	c0001	7	1294.61633	170.015807	184.945190	0.085714	24.0	7
2	c0002	2	88.58400	24.495200	44.292000	0.100000	8.0	2
3	c0003	4	342.34400	-34.284237	85.586000	0.150000	7.0	4
4	c0004	3	231.30800	78.524400	77.102667	0.066667	8.0	3

Predictive Modeling

A logistic regression model was built to predict whether a customer would buy a popular product.
The model achieved high overall accuracy and ROC-AUC.
However, class imbalance caused poor performance for the minority class.



Conclusions & Business Insights

Key conclusions:

A small group of customers generates most revenue
Sales show clear trends and seasonality
Large discounts reduce profitability
Customer behavior is more important than demographics

Data-driven analysis can improve marketing, forecasting,
and customer management decisions.